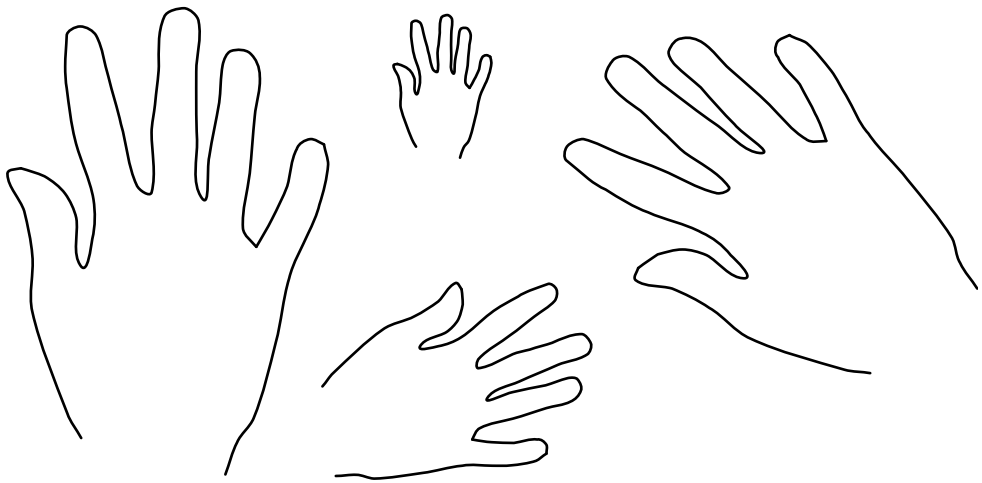
Shape & Statistical Analysis of Shape

References: see slides

Motivation

What is shape? (D.G. Kendall [Dryden & Mardia, 1998]):

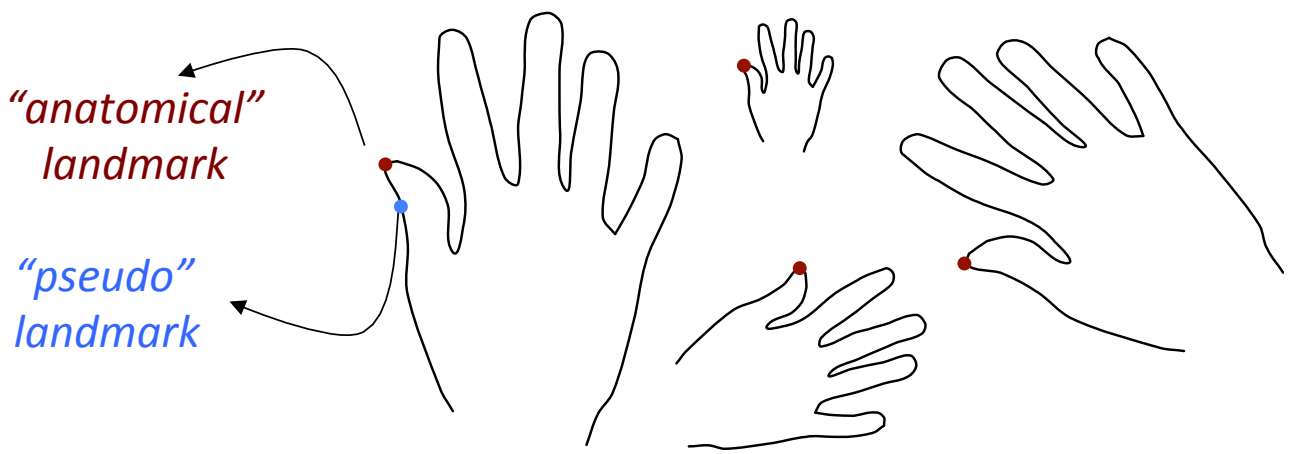
Definition (shape): *"Shape is all the geometrical information that remains when location, scale and rotational effects are filtered out from an object".*



I.L. Dryden and K.V. Mardia, "Statistical Shape Analysis", John Wiley & Sons, 1998

Motivation

How to describe shape? Common approach: use **landmarks**!



Definition (landmark): *A landmark is a "point of correspondence" on each object that matches between and within populations.*

Motivation

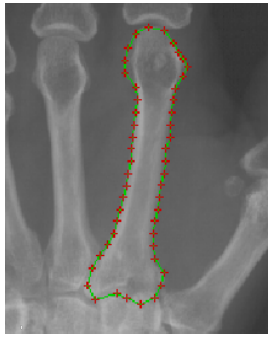
Data augmentation: we label corresponding points on each shape

Types of landmarks [Dryden & Mardia, 1998]:

(1) Application dependent (*e.g.*, medically relevant landmarks)

(2) Application independent (*e.g.*, highest/lowest points)

(3) Points that can be interpolated from (1) and (2), *e.g.*, point at equal distance around boundary (*e.g.*, the "pseudo" landmarks of previous slide)



Courtesy of M.B. Stegmann
http://www2.imm.dtu.dk/~aam/main/node12.html

I.L. Dryden and K.V. Mardia, "Statistical Shape Analysis", John Wiley & Sons, 1998

Shape space

Filtering out rotation, scale and translation leads to the concept of a "shape space".

> ***Definition (shape space):*** *The shape space is the set of all possible shapes of the object in question.*

For a more formal definition, see [Stegmann & Gomez, 2002].

Shape space

Problem: Models were allowed to deform, yet NOT tailored to the variability of the class → non-specific models

Model: **Yes** Real Life: **No**

In [Cootes et al., 1994], Cootes *et al.* state that

> *"… a model should only be able to deform in ways characteristic of the class of objects it represents"*

T.F. Cootes, C.J. Taylor, D.H. Cooper and K. Graham, "Active Shape Models – Their Training and Application", Computer Vision and Image Understanding 61(1), pp. 38-59, 1995

Point distribution model (PDM)

The "Point distribution model" (PDM) is a means for Cootes *et al.*'s [Cootes et al., 1995] "Active Shape Model"

Problem:

Mechanisms that give rise to variability are not sufficiently understood!

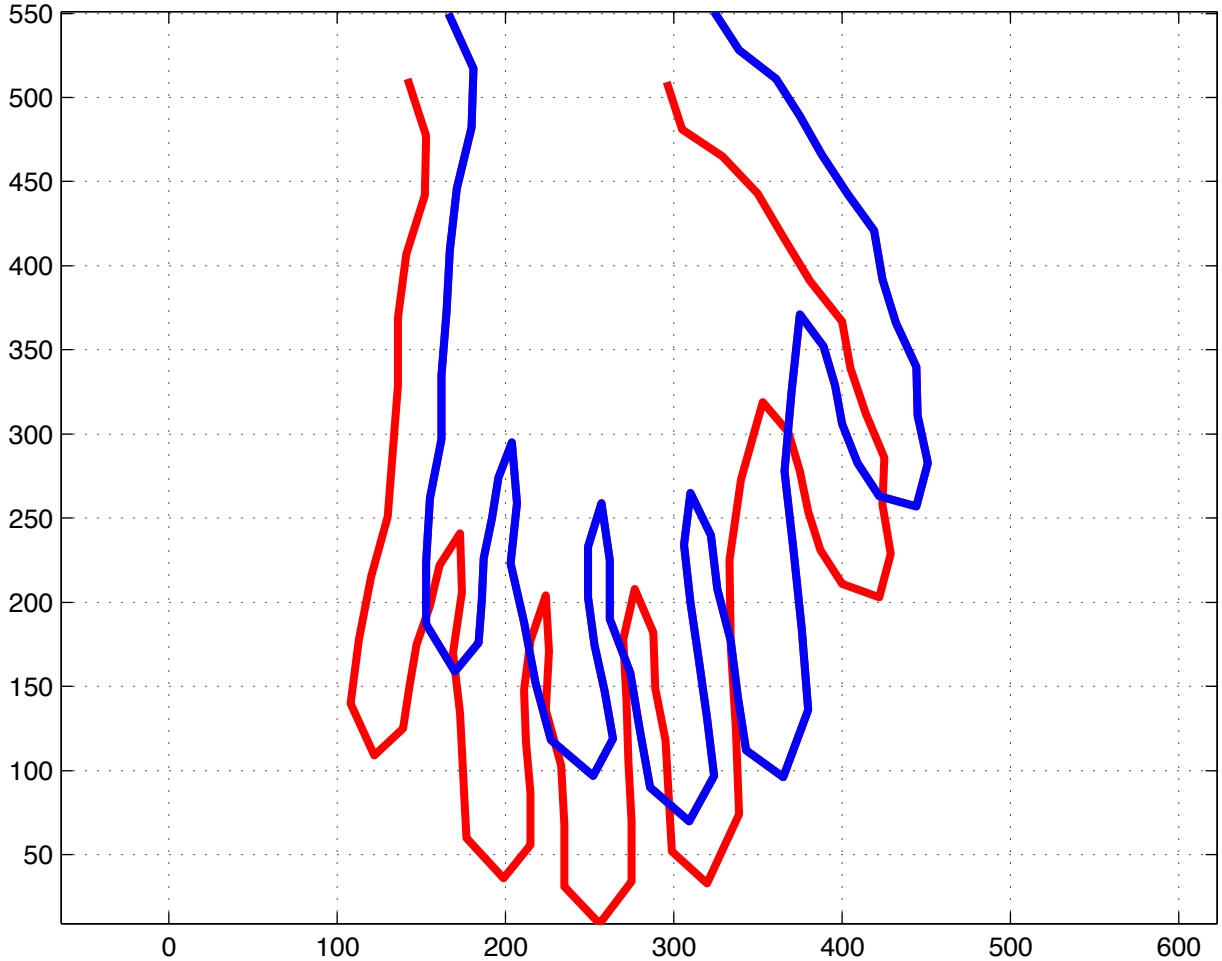
→ **Alternative**: "Learn" from existing data!

However, we do NOT want to compromise robustness for variability, which is typically the case with unspecific models!

T.F. Cootes, C.J. Taylor, D.H. Cooper and K. Graham, "Active Shape Models – Their Training and Application", Computer Vision and Image Understanding 61(1), pp. 38-59, 1995

Point distribution model (PDM)

Construction of the “shape space”



How do we map into the “shape space”?

Point distribution model (PDM)

Aligning TWO training shapes – Procrustes Analysis

Example (aligning two shapes): consider **translation only**

We need to **minimize**

$$E = \sum_{i=1}^{n} (\mathbf{x}'_i - \mathbf{T_t}(\mathbf{x}_i))^\top \mathbf{W}_i (\mathbf{x}'_i - \mathbf{T_t}(\mathbf{x}_i))$$ // n = # landmarks

with

$$\mathbf{T_t}(\mathbf{x}) = \mathbf{x} + \begin{pmatrix} t_x \\ t_y \end{pmatrix}$$ // translation

with respect to $\mathbf{t} = (t_x, t_y)$. In the <u>unweighted</u> ($\mathbf{W}_i = \mathbf{I}$) case:

$$\boxed{\hat{\mathbf{t}} = \overline{\mathbf{x}'} - \overline{\mathbf{x}}}$$

Simple to derive; just set $\mathbf{W}_i = \mathbf{I}$, take derivative w.r.t. $\mathbf{t}$ and set to 0

Point Distribution Models (PDM)

Aligning TWO training shapes – Procrustes Analysis

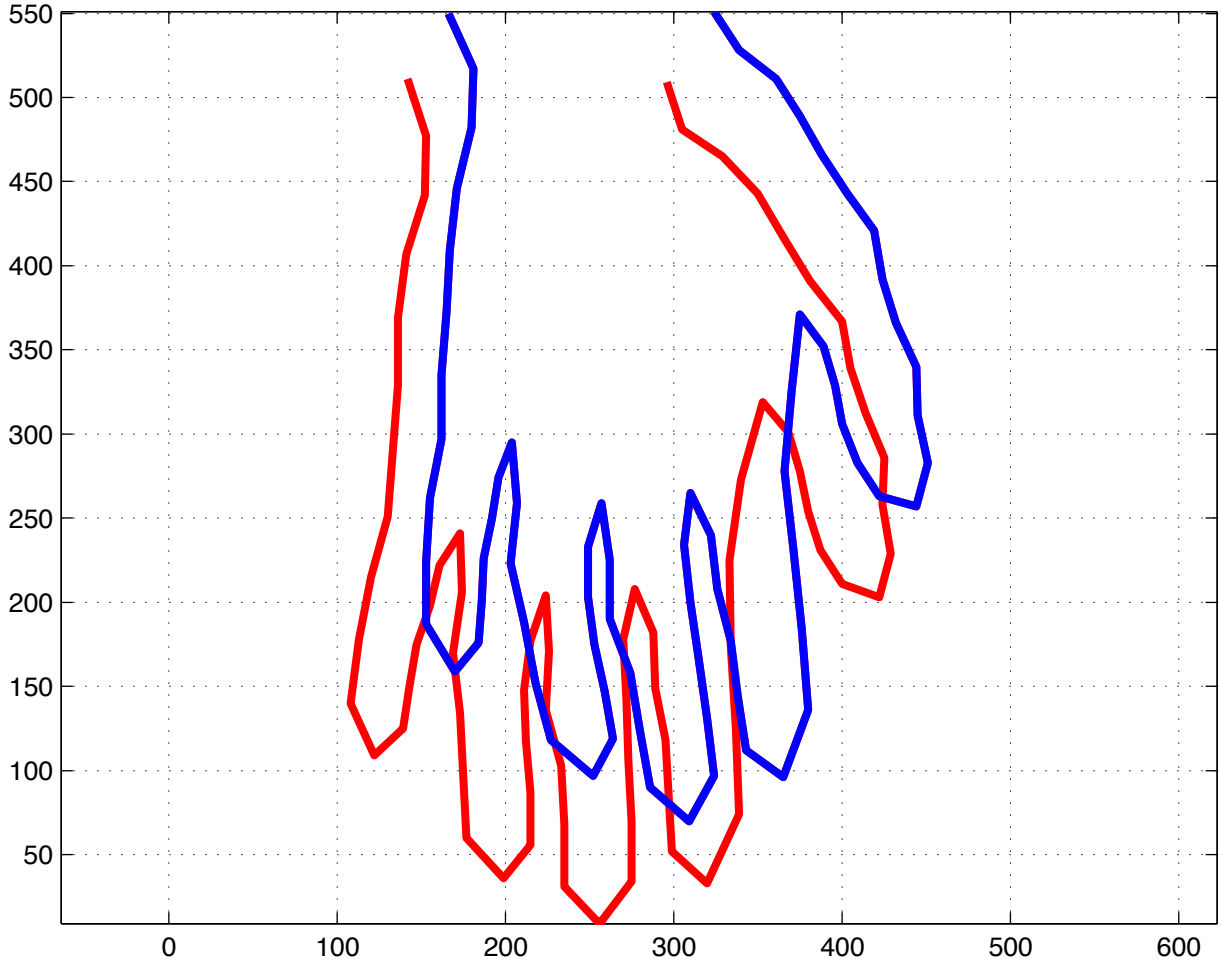
Full procrustes analysis: see MATLAB example!

(in MATLAB: >> `help procrustes`)

```
>> load cc;
>> plot(cc(:,1),cc(:,2),'r','Linewidth',3);
>> y1 = x + ones(64,2); % translation
>> y2 = 2*x; % scaling
>> y3 = 2(x + ones(64,2); % translation + scaling
>> [~,~,T1] = procrustes(x,y1);
```

Point distribution model (PDM)

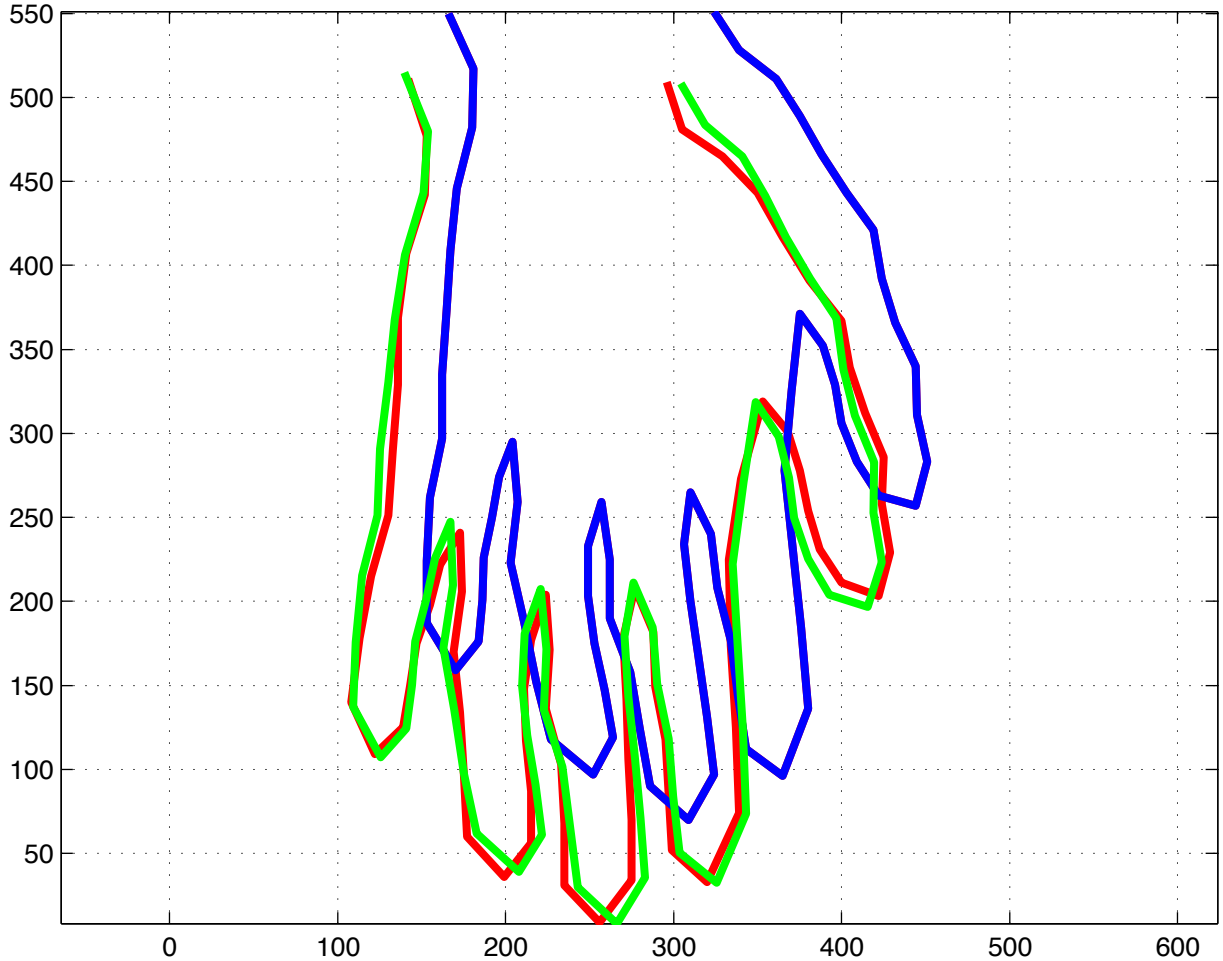
Construction of the “shape space”



Original shape (S) / Translated (T)

Point distribution model (PDM)

Construction of the “shape space”



Original shape (S) / Translated (T) / T after (full) “procrustes”

Point distribution model (PDM)

Aligning **ALL** training shapes – Procrustes Analysis

Now, to align N instances of shapes:

(1) Align (N-1) shapes with <u>first shape</u> (without loss of generality)

(2) Then, **repeat**

  (1) Calculate the mean shape from the N aligned shapes

  (2) Normalize(*) orientation, scale and origin of mean shape

  (3) Re-align all n shapes to mean shape

  until convergence

(*) Normalization could be: scale, rotate & translate mean shape to match first shape

(otherwise, problem is ill-conditioned!)

What about convergence?:

E.g., check the average difference w.r.t. aligning the samples to the mean!

No formal proof of convergence!

Point Distribution Models (PDM)

Shape statistics: Assumptions

Idea: We could model the distribution of the coordinates

*(**Remember:** we have N points in 2n-dimensional space)*

Assumption 1:

The N points lie in a part of the 2n-dimensional space and form the "allowable space domain".

Assumption 2:

The form of the point cloud is approximately ellipsoidal.

Point distribution model (PDM)

Shape statistics (1)

Let's compute the "mean" shape first:

$$\bar{\mathbf{x}} = \frac{1}{N}\sum_{i=1}^{N}\mathbf{x}_i$$

We abuse notation and consider the $\mathbf{x}_i$ as the **(already) aligned** samples!

We can then compute the <u>deviation w.r.t. the mean shape</u>

$$d\mathbf{x}_i = \mathbf{x}_i - \bar{\mathbf{x}}$$

and compute the sample covariance as

$$\mathbf{S} = \frac{1}{N}\sum_{i=1}^{N} d\mathbf{x}_i d\mathbf{x}_i^\top$$

Point distribution model (PDM)

Shape statistics (2)

Finally, PCA is used to get the **principal axis of the ellipsoid**:

$$\mathbf{S}\mathbf{p}_k = \lambda_k \mathbf{p}_k \qquad \lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_{2n}$$

(these are the 2n eigenvalues of the sample covariance matrix **S**)

As it is standard in PCA, we retain t components and then display the variations (along the first t principal modes) as:

$$\hat{\mathbf{x}} = \overline{\mathbf{x}} + \underbrace{[\mathbf{p}_1, \ldots, \mathbf{p}_t]}_{\text{eigenvectors}} \underbrace{(b_1, \ldots, b_k)}_{\text{weights}}{}^{\top}$$

$$b_k = \sqrt{\lambda_k}$$

By vary the weights → "synthesize" samples in shape space!

Point distribution model (PDM)

Shape statistics (3) - Implementation issues

What about the case when we have $N << 2n$ **training shapes?**

Reference: see Appendix B of [Cootes *et al*., 1995]

Point distribution model (PDM)

Shape statistics (4)

Data generation example: variation along the **1st** principal mode

$$\hat{\mathbf{x}}^{(2)} = \overline{\mathbf{x}} - 3\mathbf{p}_1\sqrt{\lambda_1} \qquad \hat{\mathbf{x}}^{(1)} = \overline{\mathbf{x}} + 3\mathbf{p}_1\sqrt{\lambda_1}$$

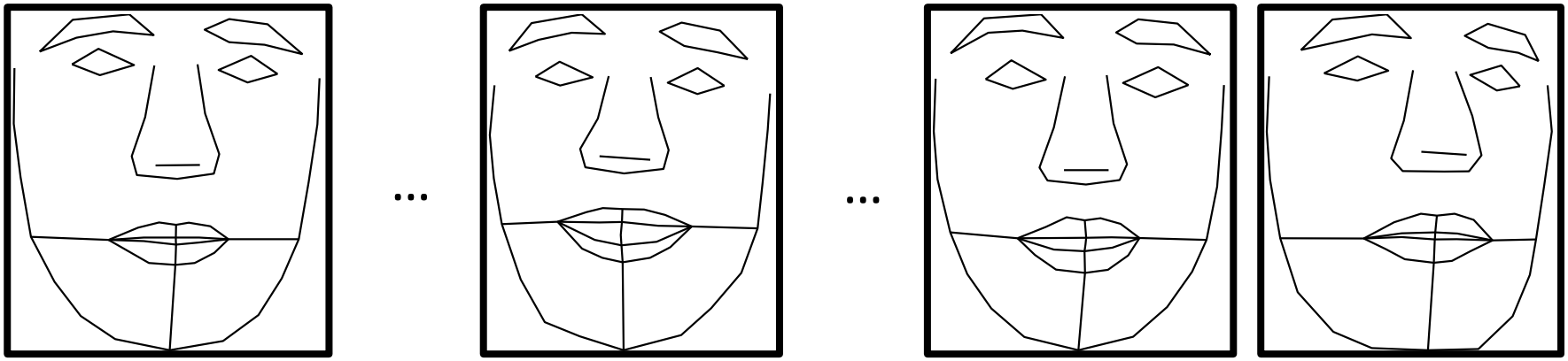
"from one extreme to the other"

(see examples on the next slides)

Point distribution model (PDM)

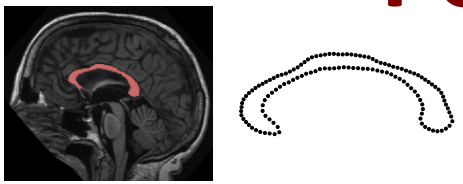
Examples – Part I (Faces)

Dataset: (landmark-based) face representations



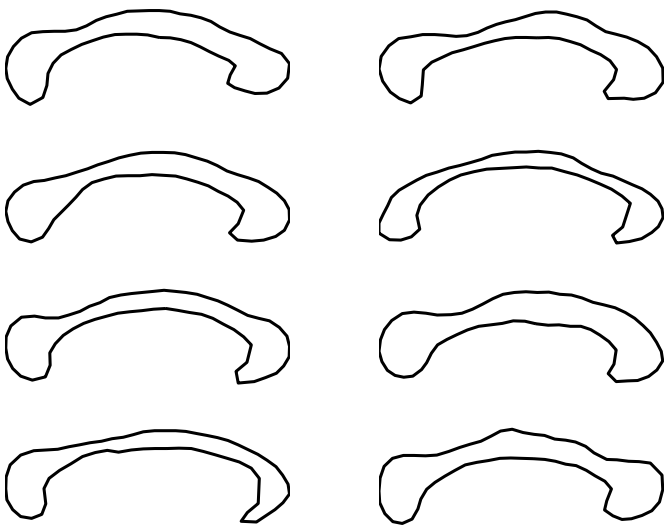
Point distribution model (PDM)

Examples – Part II (Corpus callosa)

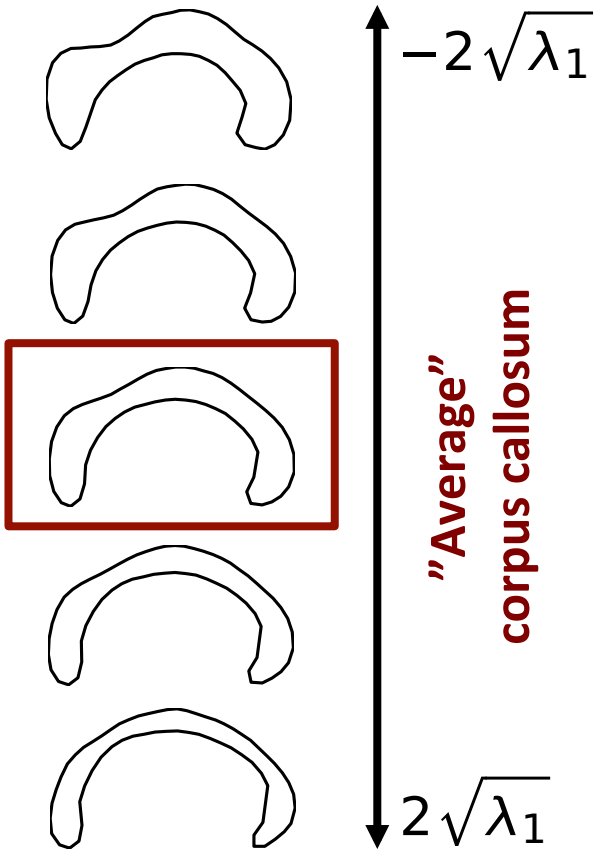


Question: What is the major mode of variation for the corpus callosum (CC) with respect to the process of aging?

Data source: Tom Fletcher



Some exemplary CC shapes (8/32)

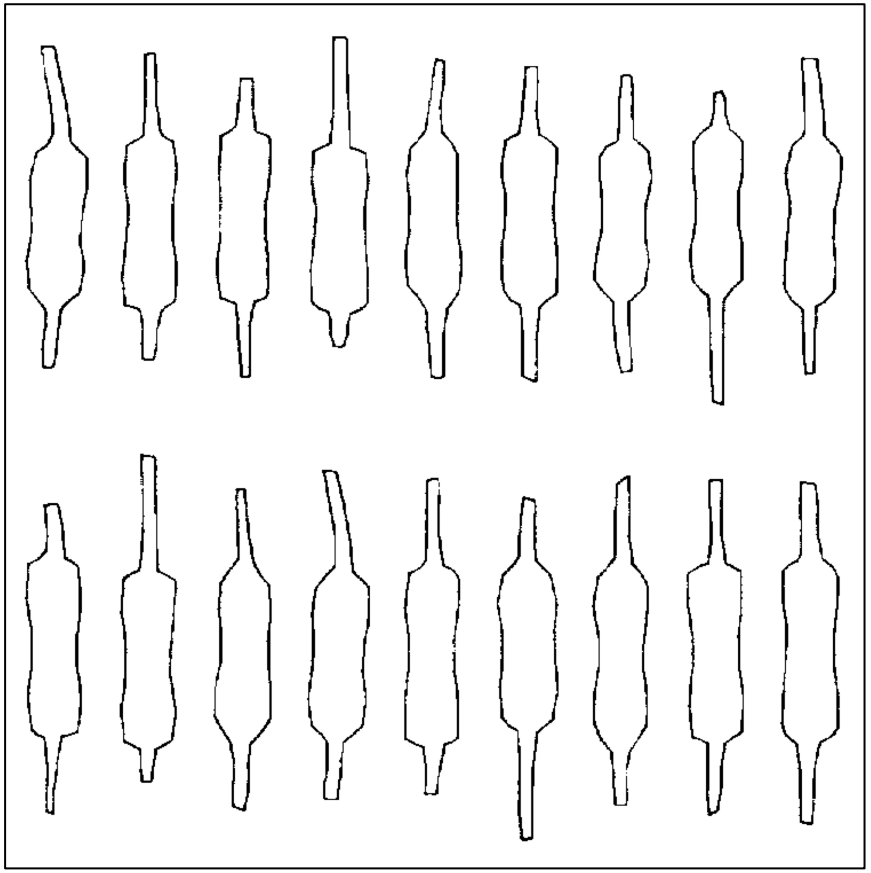


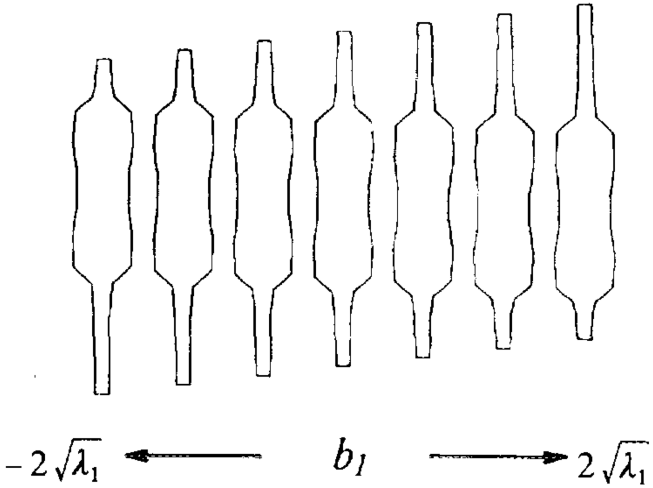
Observation: Decline in volume!

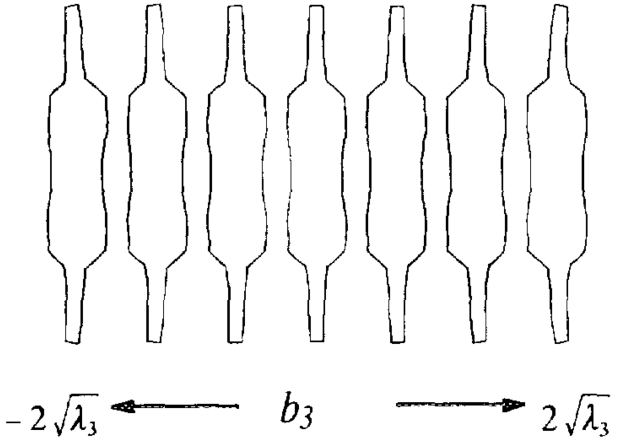
Point distribution model (PDM)

More examples (from Cootes *et al.*'s paper)

Dataset: Resistors



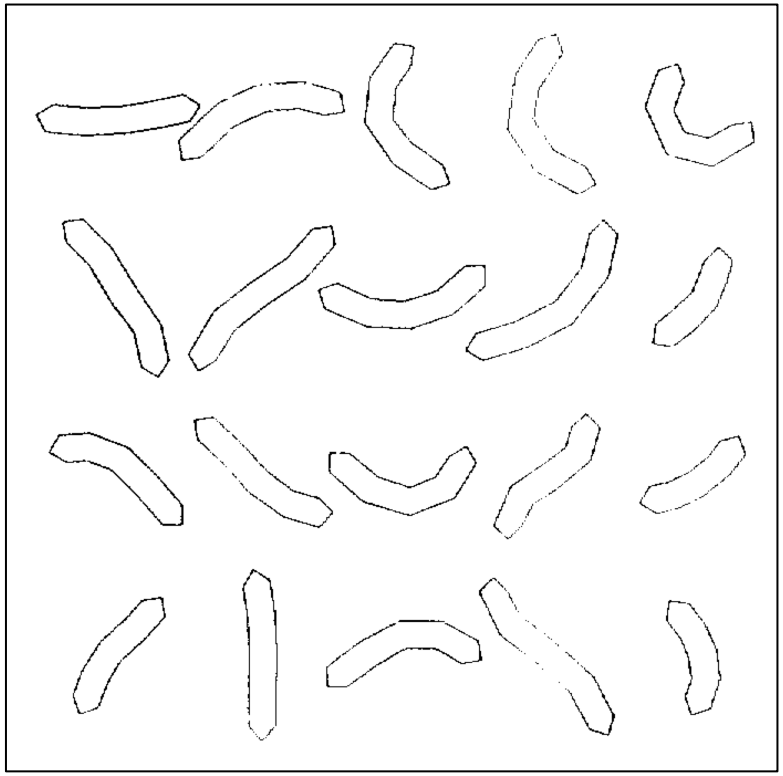


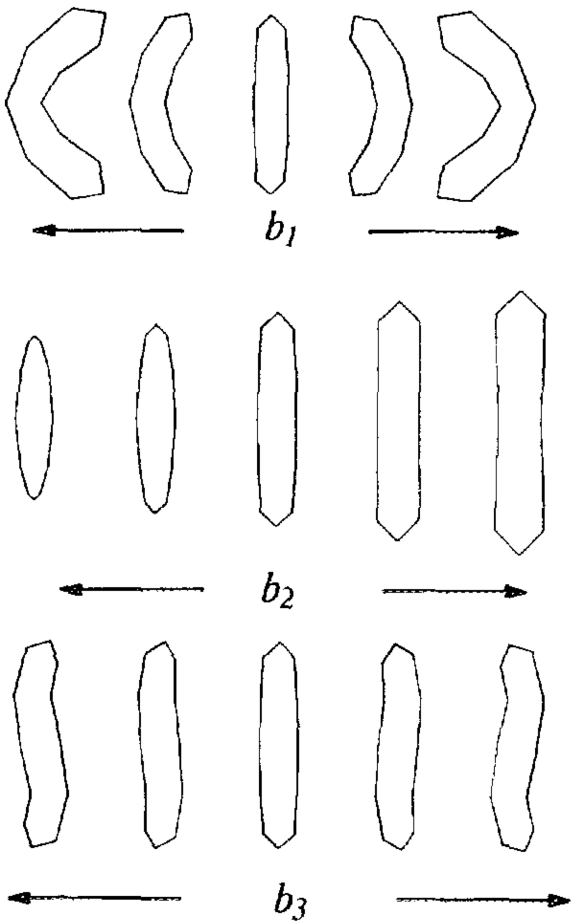


Point distribution model (PDM)

More examples (from Cootes *et al.*'s paper)

Dataset: "Worm" shapes





References

[Dryden & Mardia, 1998] I.L. Dryden and K.V. Mardia, “Statistical Shape Analysis”, John Wiley & Sons, 1998

[Stegmann & Gomez, 2002] M. Stegmann and D.G. Gomez, “A Brief Introduction to Statistical Shape Analysis”, Course Handout 2002 (available online)

[Cootes et al., 1995] T.F. Cootes, C.J. Taylor, D.H. Cooper and K. Graham, “Active Shape Models – Their Training and Application”, Computer Vision and Image Understanding 61(1), pp. 38-59, 1995

[Cootes and Taylor, 2008] T.F. Cootes and C.J. Taylor, “Statistical Models for Appearance for Computer Vision” (available online)